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GRADE A+**
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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: Importing business datasets from flat files & MySQL and connecting with any other sources into Power BI

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Experiment No.: 1

Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Github Link : <https://github.com/Lakhan07AU/Bussiness-Analytics-Projects/tree/main/Experiment-1>

Problem Statement

You are a Data Analyst in a retail company. Sales data is stored in Excel files, customer data in CSV files, and product information in another source. Management wants a centralized reporting system. Import data from multiple sources into Power BI, establish the required connections, and prepare the data for business reporting.

Step 1: Data Collection

The data required for the business analysis was collected from multiple sources.

- Sales data for 2020 and 2021 was collected for analyzing sales transactions.
- Customer information was obtained from the Customer Lookup dataset.
- Product information was obtained from the Product Lookup dataset.
- A Calendar table was used for performing date-based analysis.
- The Product Lookup dataset was also stored and managed using MySQL Workbench.

Step 2: Product Data Management Using MySQL

- Opened MySQL Workbench to manage the Product Lookup dataset.
- Created a database for storing product-related information.
- Imported the Product Lookup dataset into the MySQL database.
- Verified the product records and important fields such as:
 - ProductKey
 - ProductName
 - ProductSKU
 - ProductColor
 - ProductCost
 - ProductPrice
 - ProductSize
- Used the MySQL database as one of the data sources for the Power BI project.

Step 3: Data Import into Power BI

- Opened Power BI Desktop.
- Imported the Sales Data 2020 and Sales Data 2021.
- Imported the Customer Lookup dataset.

- Connected Power BI with the MySQL database through the MySQL Workbench environment to obtain the Product Lookup data.
- Imported/loaded the required tables into Power BI.

Step 4: Data Cleaning and Transformation

- Opened the data in Power Query Editor.
- Checked the datasets for missing and duplicate values.
- Verified column names and data types.
- Changed data types where required.
- Removed unnecessary data.
- Ensured that key columns such as CustomerKey, ProductKey, and Date were correctly formatted.
- Prepared the datasets for data modeling and analysis.

Step 5: Data Modeling

The different datasets were connected to create a proper data model.

- Connected Product Lookup with Sales Data using ProductKey.
- Connected Customer Lookup with Sales Data using CustomerKey.
- Connected Calendar with Sales Data using the date field.
- Created appropriate relationships between lookup and sales tables.
- Verified the relationship direction and cardinality.

The resulting model allowed sales transactions to be analyzed according to products, customers, and dates.

Step 6: DAX Measures

Created DAX measures to calculate the major business KPIs.

The main measures included:

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Total Quantity
- Total Cost
- Profit Margin
- Average Order Value

For example:

Total Sales = SUM('Sales Data 2020'[Sales]) + SUM('Sales Data 2021'[Sales])

Total Profit = [Total Sales] - [Total Cost]

Profit Margin = DIVIDE([Total Profit], [Total Sales])

Average Order Value = DIVIDE([Total Sales], [Total Orders])

Step 7: Dashboard Development

A Business Performance Dashboard was developed using Power BI.

The dashboard contains:

- Total Sales KPI
- Total Orders KPI
- Total Customers KPI
- Total Quantity KPI
- Total Profit KPI
- Monthly Sales Trend
- Product-wise Sales Analysis
- Customer Analysis by Gender

Step 8: Interactive Filters and Slicers

To make the dashboard interactive, slicers and filters were added for:

- Date
- Product Name
- Occupation
- Gender

These filters allow users to dynamically analyze business performance based on different customer, product, and time-based selections.

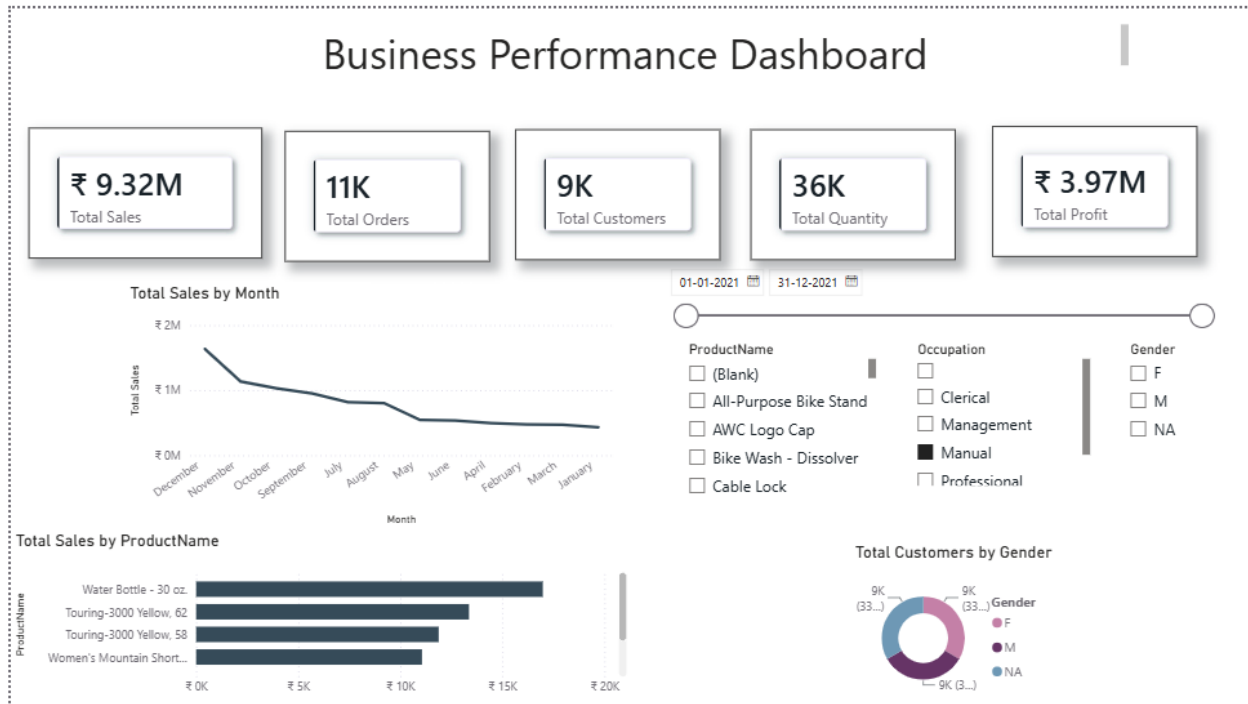
Step 9: Dashboard Formatting

- Added the title Business Performance Dashboard.
- Arranged KPI cards and charts in a structured layout.
- Applied suitable formatting to improve readability.
- Formatted sales and profit values as currency.
- Adjusted chart titles, labels, and sizes.
- Ensured that the dashboard was visually clear and easy to understand.

Step 10: Analysis and Insights

The completed dashboard was used to analyze:

- Overall sales performance.
- Overall profit generated.
- Number of orders and customers.
- Total quantity of products sold.
- Monthly sales trends.
- Products contributing to sales.
- Customer distribution by gender.
- Business performance for selected dates and customer/product categories.



Step 11: Final Dashboard

After completing data integration, transformation, modeling, DAX calculations, and visualization, the final Business Performance Dashboard was developed.

The dashboard provides management with a centralized and interactive view of business performance and helps in making data-driven decisions.

Conclusion

The Business Performance Dashboard project successfully integrated data from multiple sources, including Sales Data, Customer Lookup data, and Product Lookup data managed through MySQL Workbench. The data was cleaned, transformed, modeled, and connected in Power BI to create meaningful business insights.

DAX measures were developed to calculate important performance indicators such as Total Sales, Total Profit, Total Orders, Total Customers, and Total Quantity. Interactive charts, KPI cards, slicers, and filters were used to present the information in an easy-to-understand format.

The final dashboard provides a centralized view of business performance and helps users analyze sales trends, product performance, customer information, and profitability. Overall, the project demonstrates how MySQL and Power BI can be combined to transform raw business data into an interactive and useful decision-support dashboard.

Learning Outcomes

Through this project, I learned:

1. **Data Collection and Integration**

Learned how to work with data collected from multiple sources such as Excel, CSV, and MySQL.

2. **MySQL Database Management**

Learned how to import and manage the **Product Lookup dataset using MySQL Workbench** and use database data for analysis.

3. **Power BI Data Import**

Learned how to connect different data sources to Power BI and load the required datasets.

4. **Data Cleaning and Transformation**

Learned how to use **Power Query** to clean data, handle missing values, check duplicates, rename columns, and modify data types.

5. **Data Modeling**

Learned how to create relationships between **Sales, Customer, Product, and Calendar tables** using common keys such as CustomerKey, ProductKey, and date fields.

6. **DAX**

Learned how to create **DAX measures** for calculating business KPIs such as Total Sales, Total Profit, Total Orders, Total Customers, Total Quantity, and Profit Margin.

7. **Data Visualization**

Learned how to create KPI cards, line charts, bar charts, and donut charts to represent business information visually.

8. **Interactive Dashboard Development**

Learned how to use **slicers and filters** to make the dashboard interactive and allow users to analyze data based on different conditions.

9. **Business Analysis**

Learned how to identify sales trends, analyze product performance, understand customer distribution, and evaluate overall business performance.

10. **Data-Driven Decision Making**

Gained practical knowledge of how business intelligence tools can convert raw data into meaningful insights that support better decision-making.

